



CONSCIENCE IQ

An efficiency data layer for land-based operations

Observational pilot study · Ascent Resources · Utica Shale, Ohio

A six-week, low-disruption observational study to establish whether measurable human-readiness patterns vary across shifts and rotations — without changing staffing, workflow or existing management procedures.

THE QUESTION THE PILOT ANSWERS

At what point in a rotation does measurable readiness deterioration begin — and does it vary by day tour, night tour, or operation type?

Something you can plan against, not just a set of scores.

WHAT WE DO

Measure human-readiness factors — a brief seated reading at shift handover. **90 seconds per person** once the baseline is set

Build an efficiency data layer — how readiness varies by tour, by position in a hitch, by weather and travel

Model scenarios — a digital twin sandbox that tests operational changes before you make them

Hand it to management — Ascent interprets and decides

WHAT WE DO NOT DO

Fitness for duty — not assessed, not produced, not our purpose

Vendor ranking — contractors are not compared or scored

Automated decisions — no analysis or prediction is issued without Ascent's agreement

Added paperwork — no new procedure, permit, or reporting burden on the pad

The two outcomes we are aiming at

Efficiency

A crew in the back half of a demanding rotation is slower before it is ever unsafe. That loss is currently invisible and unpriced.

Avoidable injury

Human-factor incidents cluster in conditions that can be identified in advance and planned around.

Low commitment

A six-week pilot, not a subscription. If it does not earn its place, there is nothing to continue and nothing to cancel.

Beyond the pilot

With enough readiness data, the same measurements support an anonymised working model of the workforce — a digital twin that can be run against the operating variables which move productivity: seasonal conditions, travel disruption, schedule intensity, hitch length and crew composition. A change could then be tested before it is made.

That is a later capability, not a pilot deliverable. Six weeks establishes whether the underlying measurements hold up. Nothing in this proposal commits Ascent to anything beyond it.

THE COMMERCIAL POSITION

We supply the evidence layer. Ascent supplies the judgement.

The aim is to give you the most usable picture of your workforce for planning the next quarter — getting the most from your crews without burning them out. No analysis, prediction, or recommendation is issued without management's agreement.

The pilot in detail

Six weeks, one drilling rig and one frac spread, approximately sixty participants. Observational throughout: nothing about staffing, workflow or management procedure changes for the duration. Governed by the CIQ Scientific Technical Design Document v1.0.

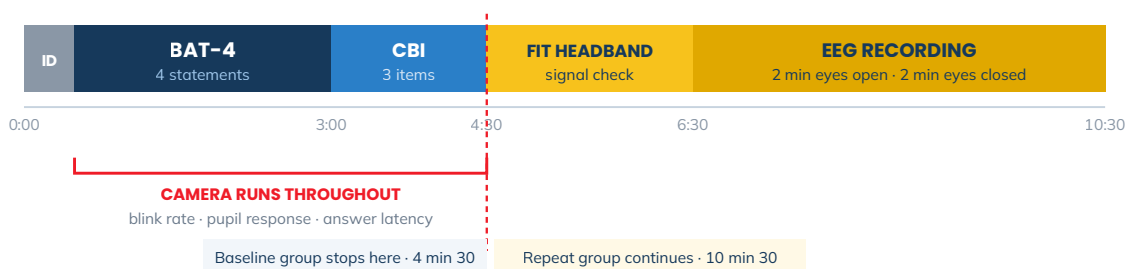
What it costs your operation

Item	Detail
Time per person	Baseline session: approximately ten minutes, once. Every session after: approximately 90 seconds
Where	Crew trailer or frac van. Never on the pad, never in a classified area
When	After the JSA and pre-tour safety meeting. We never compete with it
Worn during work	Nothing. Equipment is removed before the participant returns to the tour
New paperwork	None. No permit, procedure, or reporting obligation is created
Duration	Six weeks, covering two full crew rotations
Commitment	The pilot only. No subscription, no minimum term, no cancellation required

On timing. Only the first session takes ten minutes; it establishes the personal baseline that everything afterwards is measured against. Subsequent sessions are around 90 seconds and sit inside the handover window, which is already transitional rather than productive. It should be paid time, and that is the only operational ask in this proposal.

What one session consists of

ANATOMY OF A SINGLE ASSESSMENT



Seven questions on a tablet, then a soft headband worn while seated. The camera runs during the questions only. There is no separate test, no drill, and no physical demand.

This full sequence is the **baseline session**, taken once per participant. Repeat sessions drop the questionnaire entirely and run a short headband reading against that established baseline — approximately 90 seconds.

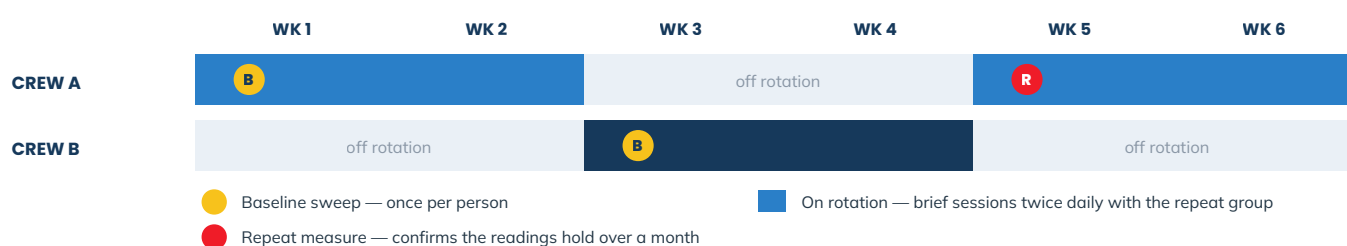
Who is measured, and how often

Group	Size	What they do
Baseline group	~60 people	One session each, spread across the first week. Provides the reference measure for everyone taking part
Repeat group	~22 volunteers	Brief sessions twice daily at handover. Provides the reliability needed to make the data usable

Why a small repeat group rather than a large single sweep. Measuring twenty-two people repeatedly produces a more reliable result than measuring two hundred and seventy people once. This is why the pilot asks for a modest number of participants rather than a large share of the workforce.

The six weeks

TWO CREW ROTATIONS, BOTH TOURS



Crews work two weeks on and two off, so six weeks covers Crew A, then Crew B, then Crew A again. The return visit in week five confirms the measurements hold steady over a month — a check that costs nothing extra and materially strengthens the result.

Staffing and footprint

Item	Detail
CIQ field team	Two operators. Handovers are twelve hours apart; one person cannot cover both
Equipment on site	Tablets and four to six headbands in charging and cleaning rotation, with a written cleaning procedure
Space required	A seat and a table in the crew trailer or frac van for approximately forty-five minutes, twice daily
Site access	Standard induction and badging. No area is entered without written HSE clearance

Data, privacy and contract workers

Ascent engages contract workers through drilling and pressure-pumping companies. Those companies will want to know what this pilot does to their people before releasing them, and their agreement is required for the pilot to run.

The position below is written so that it can be passed to them directly, or used as the basis of a conversation.

The position on contract workers and their employers

Conscience IQ does not evaluate whether a contracting company has supplied good or bad workers. The pilot studies temporary human-readiness factors that affect any person during demanding work, irrespective of experience, employer, or competence. Contract workers are treated identically to directly employed personnel throughout.

Individual results remain anonymous to the operator, to contracting company management, and to supervisors: no result any of them sees can be attached to a named person. The pilot produces no fitness-for-duty decisions, employment recommendations, contractor rankings, or grounds for removing a worker from employment.

Management receives anonymised, crew-level findings only, intended to identify operational patterns and opportunities for support. The purpose is to help Ascent and its contracting companies reduce avoidable human-factor risk while protecting worker dignity, operational continuity, and the existing supplier relationship.

Masked, not withheld. Identity is masked; the data is not. Ascent sees the individual data points themselves and can explore them for insight — what is removed is the ability to attach any point to a named person, not the point itself. Nothing of analytical value is lost, because the patterns worth acting on are patterns across crews, tours and conditions rather than facts about individuals.

How individual data is handled

Question	Answer
Are names recorded?	No. Each participant is identified by a code. The record linking codes to individuals is held offline by the CIQ field team and destroyed at the end of the study
What does management receive?	Anonymised findings at crew, tour, rotation and schedule level
Can a reading affect someone's job?	No. No result reaches a supervisor and no determination is made at handover. Nothing changes about how a shift is run
Is participation compulsory?	No. Participation is voluntary and declining carries no consequence
Is this a medical test?	No. No diagnosis, no fitness-for-duty determination, and no relationship to drug or alcohol testing

Confidentiality model

Conscience IQ holds the identity key

Each participant receives a random pilot ID — for example **CIQ-A17-042**. Assessment data is stored against that ID, never against the worker's name, employee number or payroll record.

The file linking pilot IDs to identities is held separately from the assessment database, encrypted, and accessible only to a named Conscience IQ privacy lead. It is not available to Ascent, to supervisors, or to the contracting companies, and is irreversibly separated at the agreed end of the pilot.

What Ascent sees

Anonymisation restricts who can be identified, not what can be learned. Ascent sees the anonymised data points themselves and can explore them for insight — the analytical value is undiminished, because the patterns worth acting on are patterns across crews, tours and conditions rather than facts about named individuals.

In practice that means:

Drilling versus frac — how readiness patterns differ between the two operations

Day tour versus night tour — the measured difference rather than the assumed one

Early versus late rotation — where in a hitch readiness begins to fall away

Distribution across readiness bands — the proportion of assessments in each band

Relationship with operating conditions — hours worked, sleep, heat, overtime and task type

Crew-level patterns — reported only where groups are large enough to prevent identification

Why group size matters. Removing names is not sufficient on its own. A finding attributed to a single night-tour derrickhand on a named crew identifies that person as surely as naming him. Reporting therefore suppresses groups below an agreed minimum size and avoids rare combinations of role, tour and crew.

What the pilot delivers, if the success criteria are met

Success criteria are agreed in writing before any field work begins. The deliverables below follow from meeting them.

From the pilot itself

A workforce readiness baseline — the first objective account of readiness across the operation, against which future measurement can be compared
Group-wise distribution — how readiness is distributed across drilling and frac, day and night tours, and position within a rotation
Higher-risk groups identified — which combinations of role, tour and rotation position show the greatest deterioration, reported at group level
Shift comparison within like groups — the same crew type compared across tours, so differences are attributable to the schedule rather than to the people
A validated method — evidence that the measurement holds up under field conditions, ready to carry forward

After the pilot, if Ascent elects to continue

Readiness dashboard — continuing crew-level visibility rather than a one-off study report
Digital twin simulation — scenario modelling of readiness and workload against seasonal conditions, schedule intensity, hitch length and crew composition

Available only by agreement

Two categories sit outside the pilot's default scope because they affect contract workers and their employers directly. Neither is provided during the pilot.

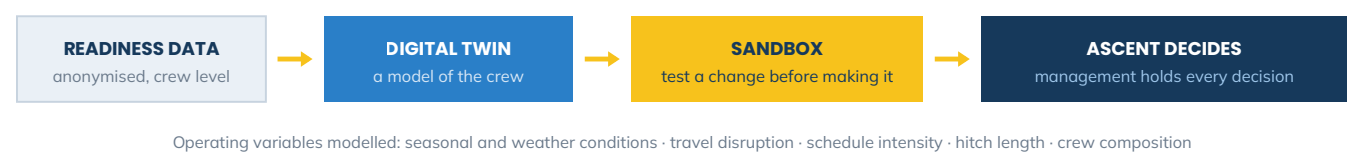
Item	Condition
Individual readiness and burnout risk scores	Released only with the written agreement of the participant, Ascent, and the relevant contracting company. The participant's own agreement is required in every case — an arrangement between two employers is not consent from the person the data concerns
Operational recommendations	Not issued during the pilot, since recommendations bear directly on contract workers and their employers. Provided afterwards only by agreement between Ascent and the contracting companies

Why these are gated. The pilot's standing is built on individual results reaching no one but the participant, and on CIQ issuing no operational prescriptions. Both hold by default. Anything beyond that is a deliberate, documented decision by every party affected — not an assumption made on their behalf.

Reporting during and after the study

During the pilot A mid-point check-in covering participation rates, data quality and any issues arising.	On completion Crew-level findings, an efficiency baseline across tours and rotations, and stated limitations.	Afterwards A case study for Ascent's review and approval, and the option of the readiness dashboard and scenario layer.
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THE DIGITAL TWIN, AS A LATER STAGE



On the digital twin. The scenario sandbox is built from data the pilot produces; it is not a pilot deliverable. Six weeks establishes the baseline in one set of conditions. Modelling a full operating cycle — summer heat through winter — requires a longer engagement, and that is a separate decision for Ascent to take on the evidence.

Proposed next steps

#	Step	Detail
1	Define success	Ascent states what the pilot must show for it to have been worth running. The protocol is built around that answer
2	Design meeting	Sixty minutes with operational leadership. Numbers, timing, crew selection
3	Contracting companies	Agreement from the drilling and pressure-pumping companies, using the position set out above. This is the longest lead item and should begin first
4	Technical session	HSE, IT and workforce representatives. Consent, site clearance and data handling. Operational leadership need not attend
5	Protocol approval	A tailored protocol is submitted for written approval before anyone travels

IN SUMMARY

Conscience IQ measures what a rotation costs a crew in efficiency, and gives Ascent the data to plan around it.

Conscience IQ · Prepared for Ascent Resources, 27 July 2026. Pilot governed by the CIQ Scientific Technical Design Document v1.0 (30 June 2026). Individual results are disclosed to the participant alone; management reporting is anonymised and at crew level. The pilot produces no medical assessment, diagnosis, or fitness-for-duty determination, and supplies no basis for employment or contracting decisions. Participant numbers are planning estimates. Digital twin scenario modelling is a subsequent-stage capability and is not a deliverable of the six-week pilot.